Tom Dror

CLIMATE CHANGE · CLOUD PHYSICS · LAND ATMOSPHERE INTERACTION

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Education

Weizmann Institute of Science *Israel*

PH.D. DEPARTMENT OF EARTH AND PLANETARY SCIENCES *2018 - 2023*

- Dissertation title: On the properties of *greenCu*: continental, organized shallow clouds
- Advisor: Prof. Ilan Koren

Weizmann Institute of Science *Israel*

MSC. DEPARTMENT OF EARTH AND PLANETARY SCIENCES *2015 - 2018*

- Dissertation title: Timescale analysis of environmental controls on Sea Spray Aerosol production over the South Pacific Subtropical Gyre
- Advisor: Prof. Ilan Koren

Hebrew University of Jerusalem *Israel*

BSC. INSTITUTE OF EARTH SCIENCE *2009 - 2012*

- Climate, Atmospheric Science and Oceanography

Professional Experience

2024-present	**CIRES Postdoctoral fellow**, Cooperative Institute for Research in Environmental Sciences (CIRES), University of Colorado Boulder & NOAA Chemical Sciences Laboratory (R/CSL9)
2023-2024	**Postdoctoral fellow**, Weizmann Institute of Science
2018-2024	**CloudCT**, Member of the CloudCT project, a unique interdisciplinary Israeli-German collaboration between the Weizmann Institute, the Technion, and ZFT, aimed at launching a formation of ten tiny satellites that will utilize medically-inspired computed tomography (CT) techniques to address critical climate questions (**link**).
2013-2015	**Project manager and environmental planner**, Ethos Architecture Planning and Environment. Managing planning teams in master and outline plans. A member of the integrating team of the *Israeli Marine Plan.*

Publications

PUBLISHED & IN PRESS

Dror, T., & Feingold, G. (2026). Amazon Forest Loss: an All-Sky Biophysical TOA Cooling Feedback. *Science* (in press).

Dror, T., Flores, J. M., & Koren, I. (2025). Global diurnal sea surface temperature variability and the role of ocean-atmosphere interactions. *Journal of Geophysical Research: Oceans*, 130(8), e2025JC022862. link.

Koren, I., **Dror, T.**, Shehter E. R., & Altaratz, O. (2025). Not as random: the stable dynamics controlling shallow convective clouds. *npj Clim Atmos Sci* 8, 43. link.

Koren, I., **Dror, T.**, Altaratz, O., & Chekroun, M. D. (2024). Cloud versus Void Chord Length Distribution (LvL) as a Measure for Cloud Field Organization. *Geophysical research letters*. link.

Dror, T., Koren, I., Liu, H., & Altaratz, O. (2023). Convective steady state in shallow cloud fields. *Physical Review Letters*, 131(13), 134201. link.

Chekroun, M. D., **Dror, T.**, Altaratz, O., & Koren, I. (2023). Equations discovery of organized cloud fields: Stochastic generator and dynamical insights. *arXiv preprint* arXiv:2304.12199. link.

Dror, T., Silverman, V., Altaratz, O., Chekroun, M. D., & Koren, I. (2022). Uncovering the Large-Scale Meteorology That Drives Continental, Shallow, Green Cumulus Through Supervised Classification. *Geophysical research letters*, 49(8), e2021GL096684. link.

Dror, T., Chekroun, M. D., Altaratz, O., & Koren, I. (2021). Deciphering organization of GOES-16 green cumulus through the empirical orthogonal function (EOF) lens. *Atmospheric chemistry and physics*, 21(16), 12261-12272. link.

Dror, T., Koren, I., Altaratz, O., & Heiblum, R. H. (2020). On the Abundance and Common Properties of Continental, Organized Shallow (Green) Clouds. *IEEE Transactions on Geoscience and Remote Sensing*. link.

Dror, T., Flores, J. M., Altaratz, O., Dagan, G., Levin, Z., Vardi, A., & Koren, I. (2020). Sensitivity of warm clouds to large particles in measured marine aerosol size distributions – a theoretical study, *Atmospheric Chemistry & Physics*, 20, 15297–15306. link.

Dror, T., Lehahn, Y., Altaratz, O., & Koren, I. (2018). Temporal-Scale Analysis of Environmental Controls on Sea Spray Aerosol Production Over the South Pacific Gyre. *Geophysical Research Letters*, 45(16), 8637-8646. link.

Chemke, R., **Dror, T.**, & Kaspi, Y. (2016). Barotropic kinetic energy and enstrophy transfers in the atmosphere. *Geophysical Research Letters*, 43(14), 7725-7734. link.

Under Review

Dror, T., & Feingold, G. (2026). Deforestation Triggers State-dependent Cloud Regimes in the Amazon. Under review in *Earth's Future*.

Chekroun, M. D., **Dror, T.**, Koren, I., & Honghu, L. (2026). Multilayered Stochastic Delay Models with Hierarchical Structures: Modeling Emergent Beat Patterns. Under review in *npj Complexity*.

Roth, H., **Dror, T.**, Altaratz, O., & Koren, I. (2026). Ternary Pattern Space of Shallow Marine Clouds. In revisions for *Journal of Geophysical Research: Atmosphere*.

Zhang. J., Painemal. D., **Dror. T.**, Lim. J.S., Sorooshian. A., & Feingold. G. (2026). Inferring processes governing cloud transition during mid-latitude marine cold-air outbreaks from satellite. Under review in *Atmospheric Chemistry & Physics*. link.

Awards & Fellowships

2024	**The CHE/PBC Fellowship Program for Outstanding Female Postdoctoral Students**, Higher Education Council	*Israel*
2024	**CIRES Postdoctoral Visiting Fellowship**, Cooperative Institute for Research in Environmental Sciences at the University of Colorado Boulder	*Boulder, CO, USA*
2023	**Next-gen Environmental Sustainability Postdoc award**, Institute for Environmental Sustainability, Weizmann Institute of Science	*Israel*
2021	**PhD Research Fellowship**, Weizmann Data Science Research Center	*Israel*
2021	**Outstanding Student and Ph.D. candidate Presentation Award**, European Geosciences Union (EGU)	*Vienna, Austria*
2018	**A Travel award**, Weizmann Women in science	*Israel*

Presentations at conferences and seminars

Invited Talks and Semniras

NOAA Boulder, Chemical Sciences Laboratory (CSL) Seminar Series, May 2026.

University of Colorado, Boulder, Boase Hydrologic Sciences and Water Resources Engineering Seminar Series, February 2026.

The Weizmann Institute of Scince, The Institute for Environmental Sustainability (IES), January 2026.

The Hebrew University of Jerusalem, Institute of Earth Sciences seminar, January 2026.

AMS annual meeting: joint session of the 17th Symposium on Aerosol-Cloud-Climate Interactions and the Second Symposium on Cloud Physics, January 2025.

Technion – Israel Institute of Technology, Environmental Engineering seminar, March 2024.

Tel Aviv University, Geophysics colloquium, February 2024.

The Hebrew University of Jerusalem, Institute of Earth Sciences seminar, January 2024.

Israel Meteorological Service, IMS colloquium, January 2024.

Batsheva de Rothschild Seminar on Cloud-Climate Interactions Across Scales, March 2023. *Green Cumulus clouds throughout the world's continents*. Eilat, Israel.

Conferences

American Geoscience Union (AGU) Fall Meeting, 2025. *Cooling Feedbacks from Amazon Deforestation: An All-Sky Biophysical Perspective*. New Orleans, LA, U.S.A.

Gordon Research Conference, Radiation and Climate, 2025. *Amazon Forest Loss: an All-Sky Biophysical Cooling Feedback*. Maine, U.S.A.

AGU Fall Meeting, 2024. *On the Stable Dynamical Machinery that Dictates Trade Cumulus Cloud Properties*. Washington, D.C., U.S.A.

AGU Fall Meeting, 2023. *Convective steady state in shallow, boundary-layer cloud fields*. San Francisco, CA, U.S.A. & online.

Workshop on Cloud Organisation and Precipitation Extremes, 2023. *Organized patterns in greenCu; continental shallow convective clouds*. Trieste, Italy.

The European Geoscience Union (EGU) General Assembly, 2023. *On the properties of greenCu: continental, organized shallow clouds*. Vienna, Austria.

International Conference on Clouds and Precipitation, 2021. *Organized shallow continental (green) Cumulus*. Pune, India (virtual).

The EGU General Assembly, 2021. *EOF analysis of Green Cumulus mesoscale organization*. Vienna, Austria & online.

AGU Fall Meeting, 2021. *On the universal properties of Green Cumulus*. New Orleans, LA, U.S.A. & online.

The EGU General Assembly, 2019. *Environmental controls on sea spray aerosol production.* Vienna, Austria.

Workshop on Eulerian vs. Lagrangian methods for cloud microphysics, 2019. *Ultra-giant CCN's effect on warm clouds.* Krakov, Poland .

Teaching Experience

2025–present	**CSL Science Social Weekly Seminar**, NOAA Chemical Sciences Laboratory, Co-organizer
2023	**Cloud physics across scales - a guided reading course**, Weizmann Institute of Science, Co-lecturer
2017	**EPScon, Student Conference on Research in Environmental, Earth and Planetary Sciences, Weizmann Institute of Science**, Organizer

Mentoring

2025–present	**Lukas Rudaitis, M.Sc. Candidate**, Member of the Supervisory and Examining Committee, Department of Physics and Atmospheric Science, Dalhousie University, Canada.
2023	**Hadar Roth**, Young Weizmann scholars, Weizmann Institute of Science.
2021	**Toviah Aaronson**, Undergraduate student, scientific project at the Weizmann Institute of Science.
2020	**Noga Liberty**, Amos de-Shalit summer school, Weizmann Institute of Science.